Prodigy Anesthesia Quick Review

Glycolysis

What type of reaction is catalyzed by phosphofructokinase?	Phosphoryl transfer *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 813-817.*
Where does the citric acid cycle take place?	The citric acid cycle (Krebs' cycle) occurs within the mitochondria of eukaryotic cells. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 813-817.*
How many turns of the Krebs cycle does it take to completely oxidize 1 glucose?	Two turns since glycolysis normally produces 2 pyruvate. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 813-817.*
What must be present for both the Krebs cycle and the electron transport system to function?	Oxygen *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 813-817.*
What is the maximum number of ATPs you can get from 1 mole of glucose?	38 *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 813-817.*
Where does the Krebs cycle and the electron transport reactions occur?	In the mitochondria *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 813-817.*
Excluding any products of glycolysis or the electron transport system, what are the final products of a single turn of the Krebs cycle?	4 NADH, 1 FADH, and 1 ATP. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 813-817.*
If the electron transport system isn't working, how many ATPs can glycolysis and the Krebs cycle produce?	4 ATP *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 813-817.*
If glycolysis is working correctly and the Krebs cycle and electron transport system are not used, what are the final products produced?	2 ATP, 2 NADH, and 2 Pyruvate *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 813-817.*

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What type of reaction is catalyzed by glyceraldehyde-3 phosphate dehydrogenase?	Phosphoryl transfer *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 813-817.*
What type of reaction is catalyzed by phosphoglucose isomerase?	Isomerization *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 813-817.*
What type of reaction is catalyzed by phosphoglycerate kinase?	Phosphoryl transfer *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 813-817.*
What is the starting product for the Kreb's cycle and what compound is necessary for its formation?	AcetylCoA which is formed from pyruvate. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 813-817.*
What type of reaction is catalyzed by hexokinase?	Phosphoryl transfer *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 813-817.*
What is oxidative phosphorylation?	Oxidative phosphorylation describes a series of electron transfers that lead to the formation of adenosine triphosphate. Oxidative phosphorylation is responsible for about 90% of all oxygen consumed in the body. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 813-817.*
What type of reaction is catalyzed by pyruvate kinase?	Phosphoryl transfer *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 813-817.*
What type of reaction is catalyzed by aldolase?	Aldol cleavage *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 813-817.*
What type of reaction is catalyzed by enolase?	Dehydration *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 813-817.*
What type of reaction is catalyzed by phosphoglycerate mutase?	phosphoryl shift *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 813-817.*

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What is a phosphoryl shift reaction?	A reaction in which a phosphoryl group is shifted from one oxygen atom to another within a molecule by a mutase. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 813-817.*
How many ATP will one NADH generate?	3 ATP *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 813-817.*
What type of reaction is catalyzed by triose phosphate isomerase?	Isomerization *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 813-817.*
What two enzyme reactions in glycolysis require ATP to function (meaning that they use up ATP)?	Hexokinase and phosphofructokinase *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 813-817.*
What is an isomerization reaction?	A reaction involving the conversion of a ketose to an aldose, or vice versa, by an isomerase. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 813-817.*
What is a phosphoryl transfer reaction?	A reaction in which a phosphoryl group is transferred from ATP to a glycolytic intermediate, or from the intermediate to ADP by a kinase. Note: dehydrogenase also is a phosphoryl transfer even though it doesn't end in kinase. *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 813-817.*
What two enzyme reactions in glycolysis are responsible for generating ATP?	Phosphoglycerate kinase and pyruvate kinase *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 813-817.*
What is a dehydration reaction?	A reaction involving the removal of water by a dehydratase. (Enolase) *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 813-817.*
How many ATP does a single FADH produce?	2 ATP *Guyton AC, & Hall, JE. Textbook of Medical Physiology. 12th ed. Philadelphia: Saunders; 2011: 813-817.*

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